Introduction to reinforcement learning

Leonardo Toffalini

2026-03-28

Outline

1. Informal introduction to RL 2
2. Setting the scene 6
3. Bellman equations 14

1. Informal introduction to RL

1.1 Characteristics of RL

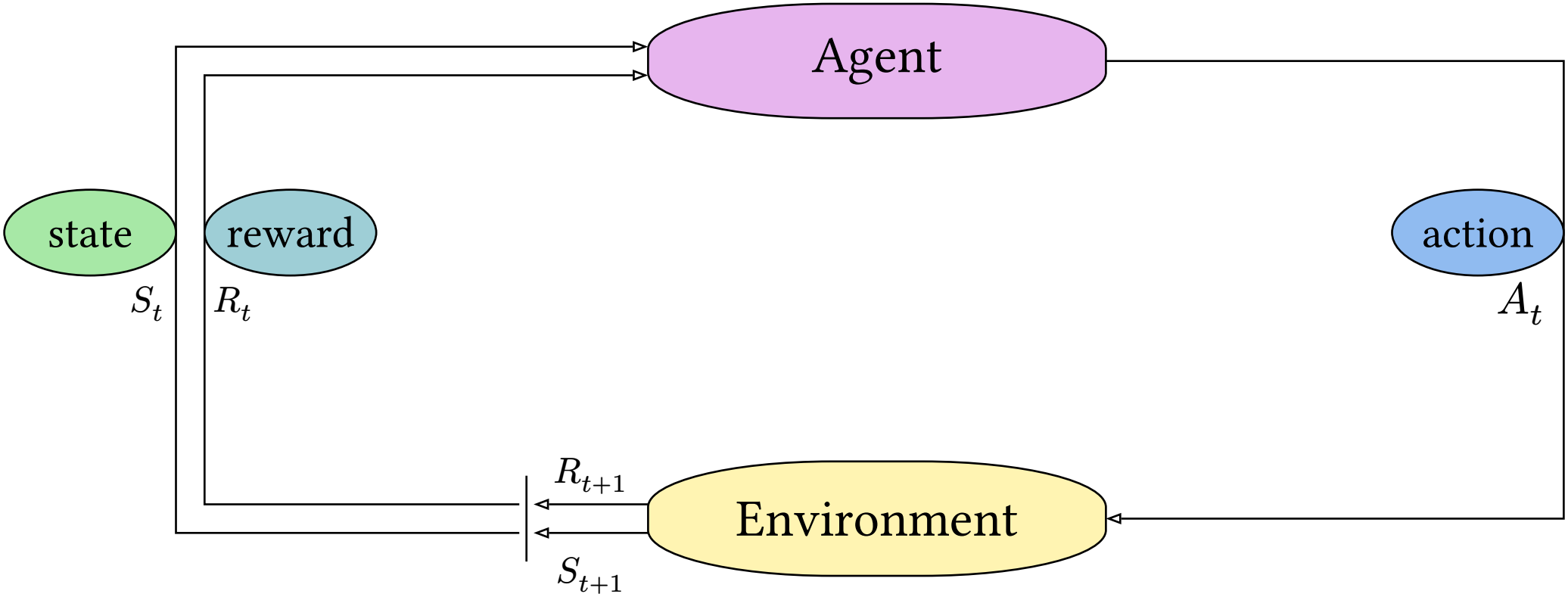
What makes RL different?

- Not supervised (learning from labeled data)
- Not unsupervised (learning patterns in unlabeled data)
- Only *reward* signal
- Feedback may be delayed
- Heavily time dependent
- The agent has influence over future data

1.2 Examples of RL

- Robotics
- Video games
- Self-driving
- Finance
- Natural language processing (recently)
- Recommendation systems
- Many more…

1.3 RL loop



2. Setting the scene

2.1 Markov decision processes

Definition 2.1.1 A Markov decision process (MDP) is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}_a, \mathcal{R}_a)$, where

- $\mathcal{S}$ is the set of all states
- $\mathcal{A}$ is the set of all possible actions
- $\mathcal{P}$ is the transition probability function

$$\mathcal{P}^a_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s, \quad A_t = a]$$

- $\mathcal{R}$ is the reward function

$$\mathcal{R}^a_s = \mathbb{E}[R_{t+1} \mid S_t = s, \quad A_t = a]$$

2.1 Markov decision processes

Definition 2.1.2 A state S_t is Markov if and only if

$$\mathbb{P}[S_{t+1} \mid S_t] = \mathbb{P}[S_{t+1} \mid S_1, \ldots, S_t].$$

2.1 Markov decision processes

Definition 2.1.2 A state S_t is Markov if and only if

$$\mathbb{P}[S_{t+1} \mid S_t] = \mathbb{P}[S_{t+1} \mid S_1, \ldots, S_t].$$

Remark

In a Markov decision process, the successor state is *solely* influenced by the current state.

That is, an MDP has no memory of previous states.

Is this a limitation?

2.2 Illustrative step in an MDP

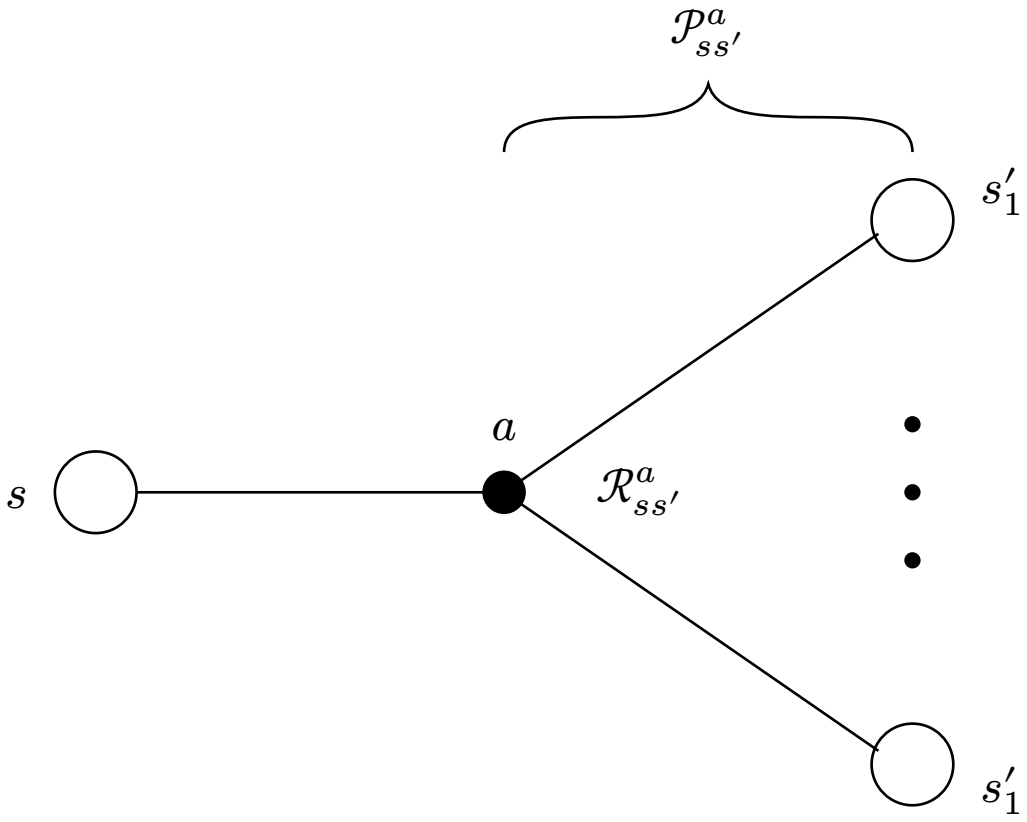


Figure 2: hollow circles = states, full circles = actions

2.3 Reward hypothesis

Definition 2.3.1 The reward hypothesis claims that all goals can be described by the maximization of expected cumulative reward.

2.3 Reward hypothesis

Definition 2.3.1 The reward hypothesis claims that all goals can be described by the maximization of expected cumulative reward.

Remark

Do you agree?

Can you find a counterexample?

Long term goals? Human values? Multiple objectives? Safety constraints?

2.4 Policy

Definition 2.4.1 A policy π is a distribution over the action space conditioned on the state space, that is

$$\pi(a|s) = \mathbb{P}[A_t = a \mid S_t = s].$$

2.4 Policy

Definition 2.4.1 A policy π is a distribution over the action space conditioned on the state space, that is

$$\pi(a|s) = \mathbb{P}[A_t = a \mid S_t = s].$$

Remark

The next action depends *only* on the current state and nothing else.

The policy may be deterministic if $\pi(a_i|s) = 1$ for some i and $\pi\big(a_j|s\big) = 0$ for any other $j \neq i$.

2.5 Value function

Definition 2.5.1 The return G_t is the total discounted reward from time step t, that is

$$G_t = R_{t+1} + \gamma R_{t+2} + \ldots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

2.5 Value function

Definition 2.5.1 The return G_t is the total discounted reward from time step t, that is

$$G_t = R_{t+1} + \gamma R_{t+2} + \ldots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

Remark

We can control the importance of *immediate* versus *future* rewards with $\gamma \in [0, 1]$. Also, $\gamma < 1$ helps with the series converging.

2.5 Value function

Definition 2.5.2 The state-value function of a state s is the expected discounted return starting from state s, that is

$$v_\pi(s) := \mathbb{E}_\pi[G_t \mid S_t = s].$$

2.5 Value function

Definition 2.5.2 The state-value function of a state s is the expected discounted return starting from state s, that is

$$v_\pi(s) := \mathbb{E}_\pi[G_t \mid S_t = s].$$

Definition 2.5.3 The action-value function of a state-action pair (s, a) is the expected discounted return starting from state s and taking action a, that is

$$q_\pi(s, a) := \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

3. Bellman equations

3.1 Bellman expectation equations

Proposition 3.1.1

$$v_\pi(s) = \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s]$$

3.1 Bellman expectation equations

Proposition 3.1.1

$$v_\pi(s) = \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s]$$

Proof.

$$\begin{aligned} v_\pi(s) &:= \mathbb{E}_\pi[G_t \mid S_t = s] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_{t+1}] \mid S_t = s] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s] \end{aligned}$$

□

3.1 Bellman expectation equations

Proposition 3.1.2

$$q_\pi(s, a) = \mathbb{E}_\pi[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]$$

3.1 Bellman expectation equations

Proposition 3.1.2

$$q_\pi(s,a) = \mathbb{E}_\pi[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]$$

Proof.

$$\begin{aligned}
q_\pi(s,a) &:= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_{t+1}, A_{t+1}] \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]
\end{aligned}$$

□

3.1 Bellman expectation equations

Proposition 3.1.3

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$

3.1 Bellman expectation equations

Proposition 3.1.3

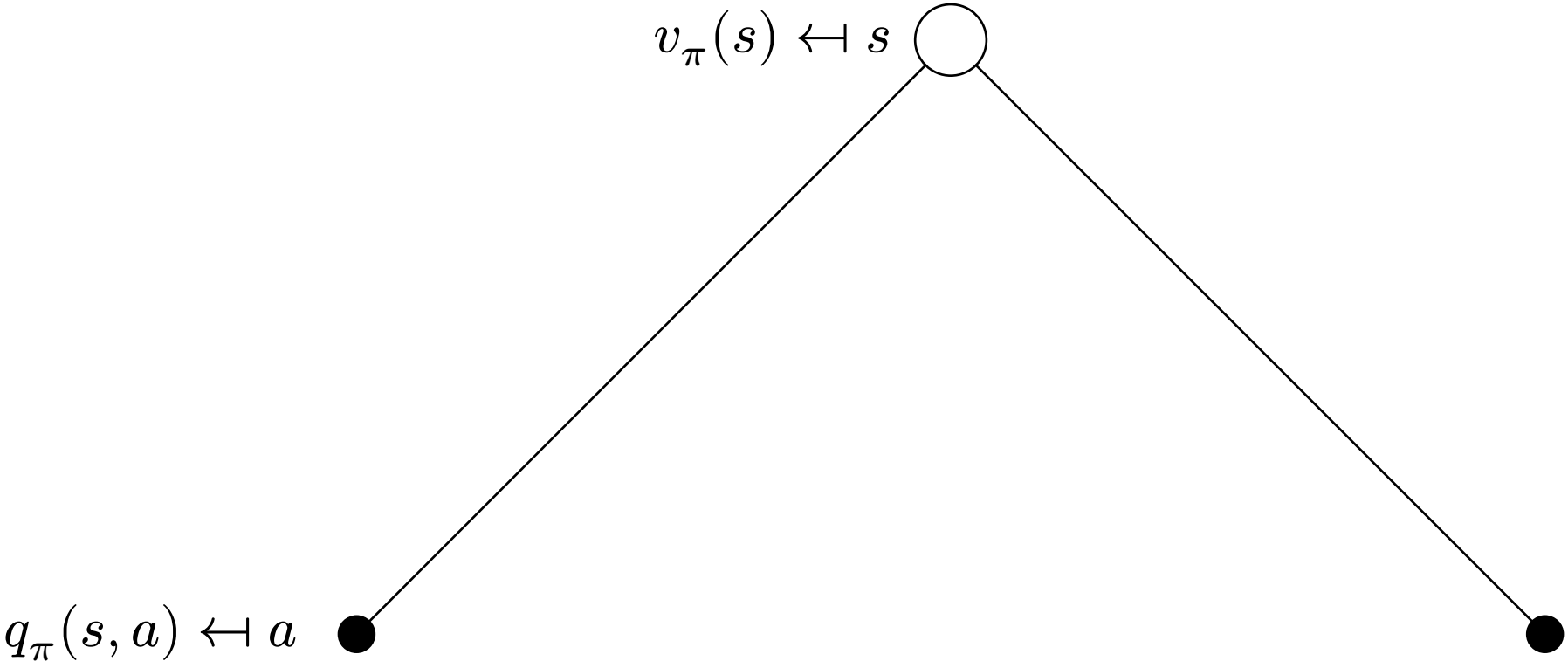
$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a)$$

Proof.

$$\begin{aligned} v_\pi(s) &:= \mathbb{E}_\pi[G_t \mid S_t = s] \\ &= \mathbb{E}_\pi[\mathbb{E}_\pi[G_t \mid S_t, A_t] \mid S_t = s] \\ &= \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s, a) \end{aligned}$$

□

3.2 Backup diagram



3.3 Bellman expectation equations

Proposition 3.3.1

$$q_\pi(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$

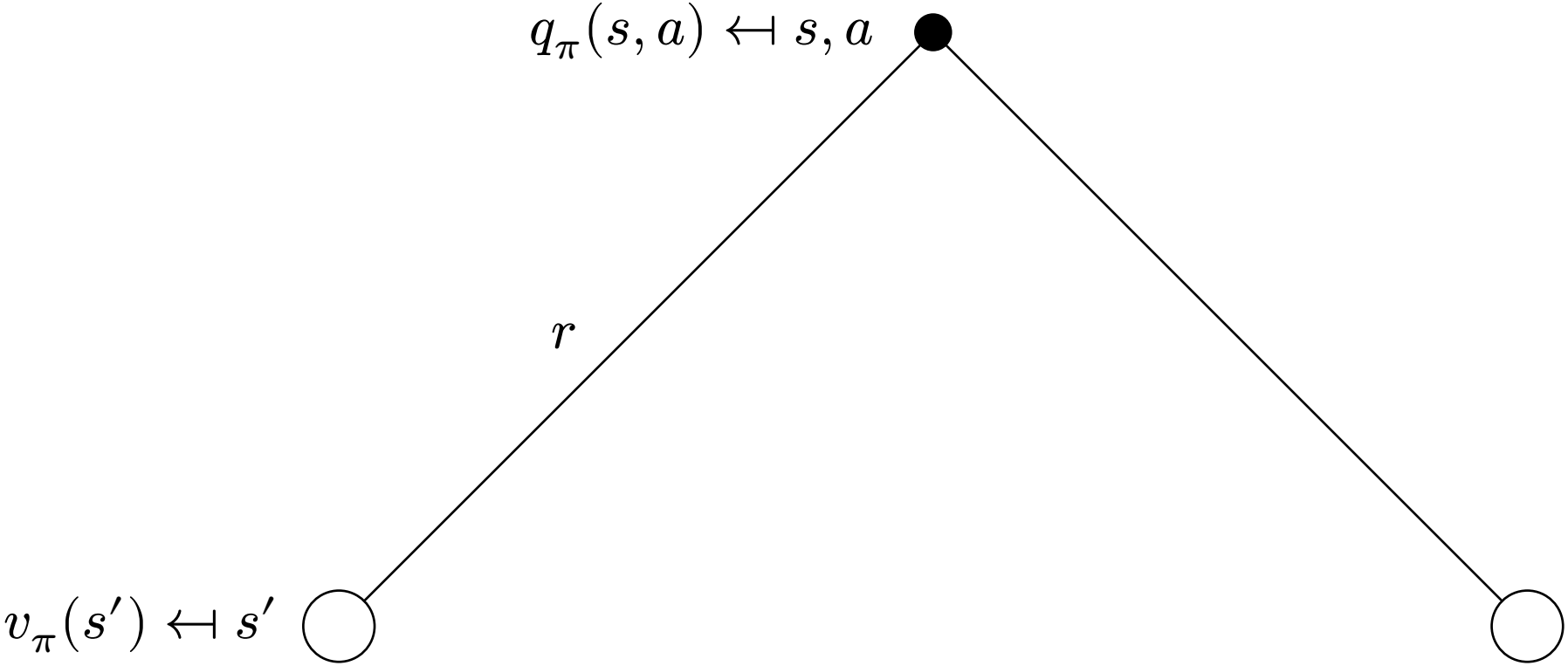
3.3 Bellman expectation equations

Proof.

$$
\begin{aligned}
q_\pi(s,a) &:= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s] \mid S_t = s, A_t = a] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s, A_t = a] \\
&= \mathcal{R}_s^a + \gamma \mathbb{E}_\pi[v_\pi(S_{t+1}) \mid S_t = s, A_t = a] \\
&= \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')
\end{aligned}
$$

$\square$

3.4 Backup diagram



3.5 Bellman expectation equations

Proposition 3.5.1

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in S} \mathcal{P}_{ss'}^a v_\pi(s') \right)$$

3.5 Bellman expectation equations

Proof.

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \underbrace{q_\pi(s,a)}$$

$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$

$$\implies v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in S} \mathcal{P}_{ss'}^a v_\pi(s') \right)$$

$\square$

3.5 Bellman expectation equations

Proposition 3.5.2

$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s',a')$$

3.5 Bellman expectation equations

Proof.

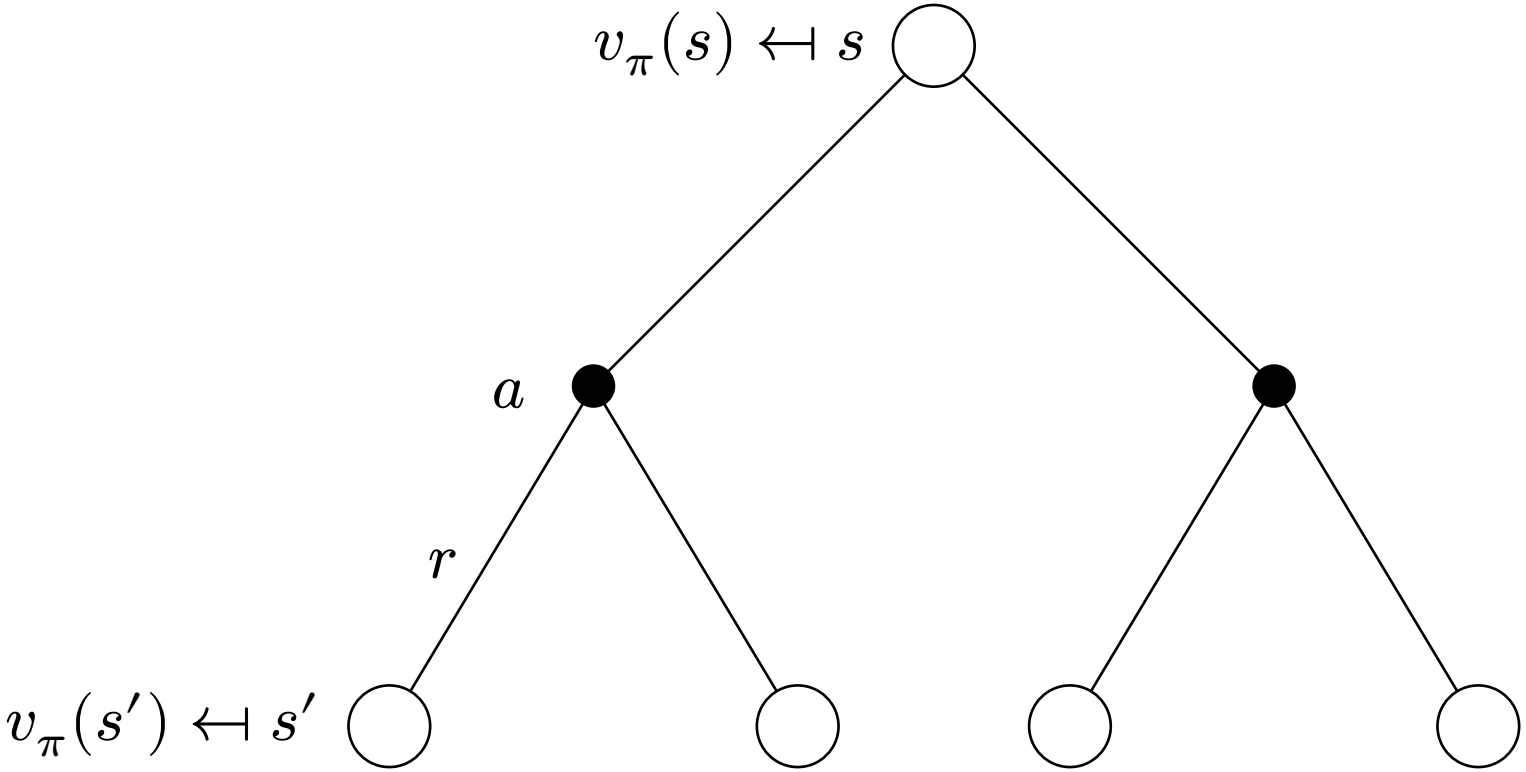
$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \underbrace{v_\pi(s')}$$

$$v_\pi(s') = \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s',a')$$

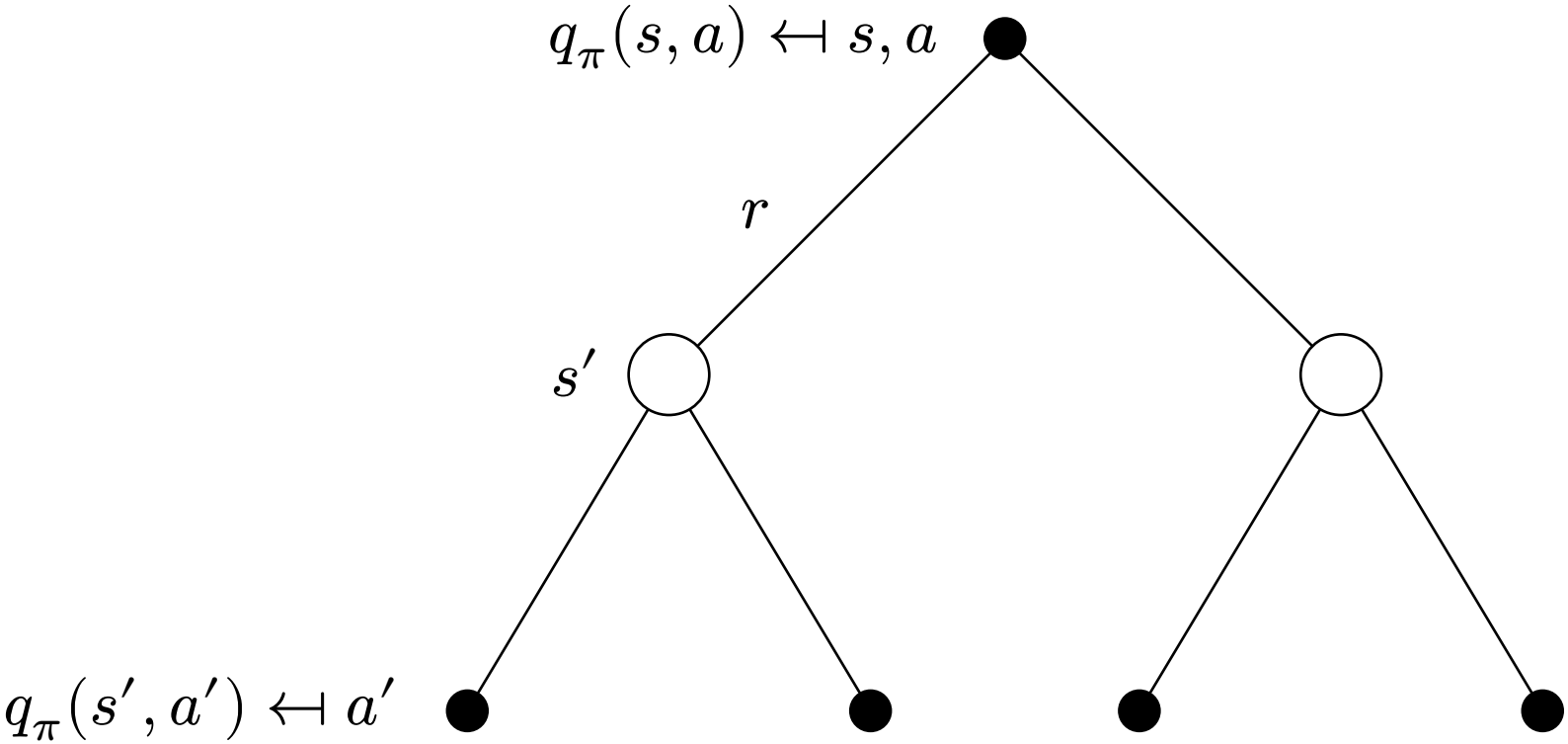
$$\implies q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s',a')$$

$\square$

3.6 Backup diagram



3.6 Backup diagram



3.7 Bellman optimality equations

Definition 3.7.1 The optimal state-value function is defined as

$$v_*(s) = \max_\pi v_\pi(s)$$

Definition 3.7.2 The optimal action-value function is defined as

$$q_*(s, a) = \max_\pi q_\pi(s, a).$$

3.7 Bellman optimality equations

Definition 3.7.3 A policy π^* is optimal if for all other policies π

$$v_{\pi^*}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

3.7 Bellman optimality equations

Definition 3.7.3 A policy π^* is optimal if for all other policies π

$$v_{\pi^*}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

Remark

That is, π^* is optimal if the value function induced by following π^* is optimal.

3.7 Bellman optimality equations

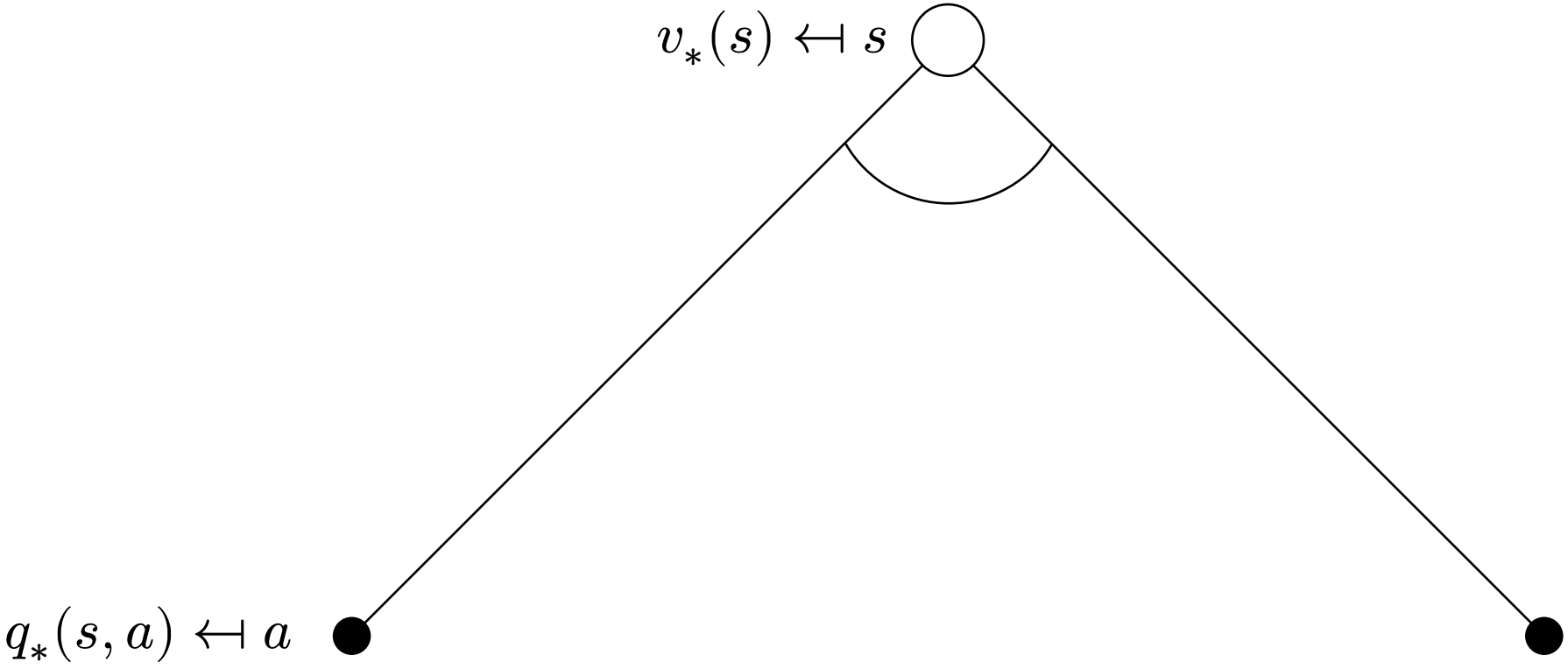
Proposition 3.7.4

$$v_*(s) = \max_a q_*(s, a)$$

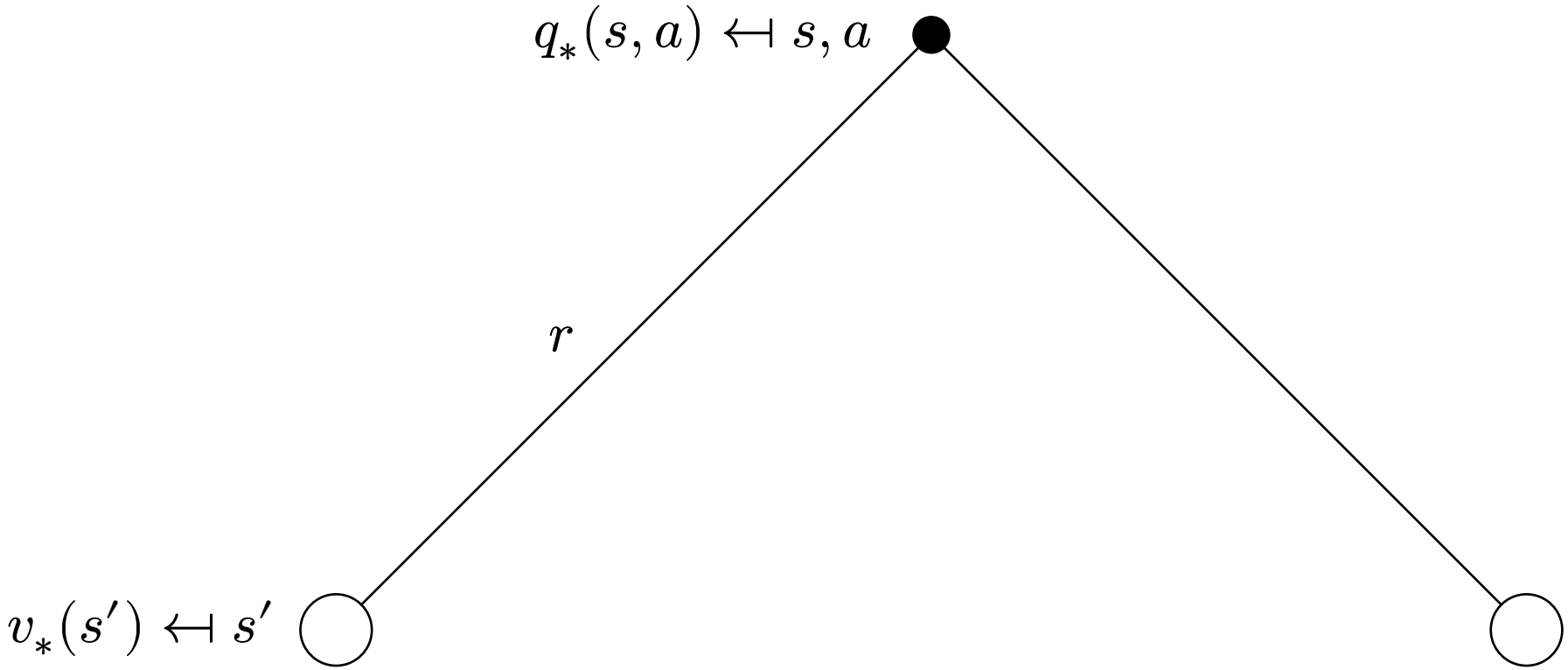
Proposition 3.7.5

$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

3.8 Backup diagrams



3.8 Backup diagrams



3.9 Bellman optimality equations

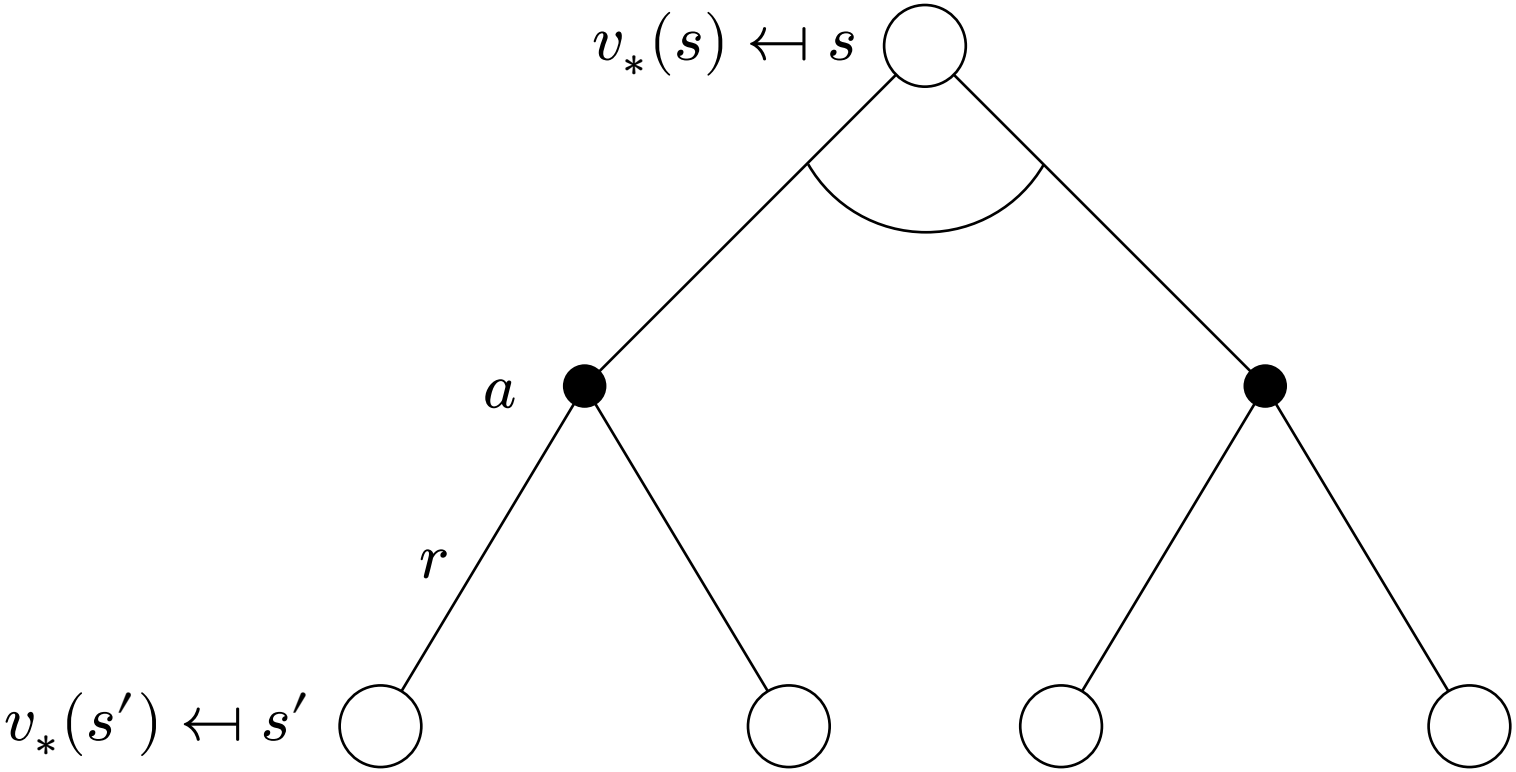
Proposition 3.9.1

$$v_*(s) = \max_a \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s') \right)$$

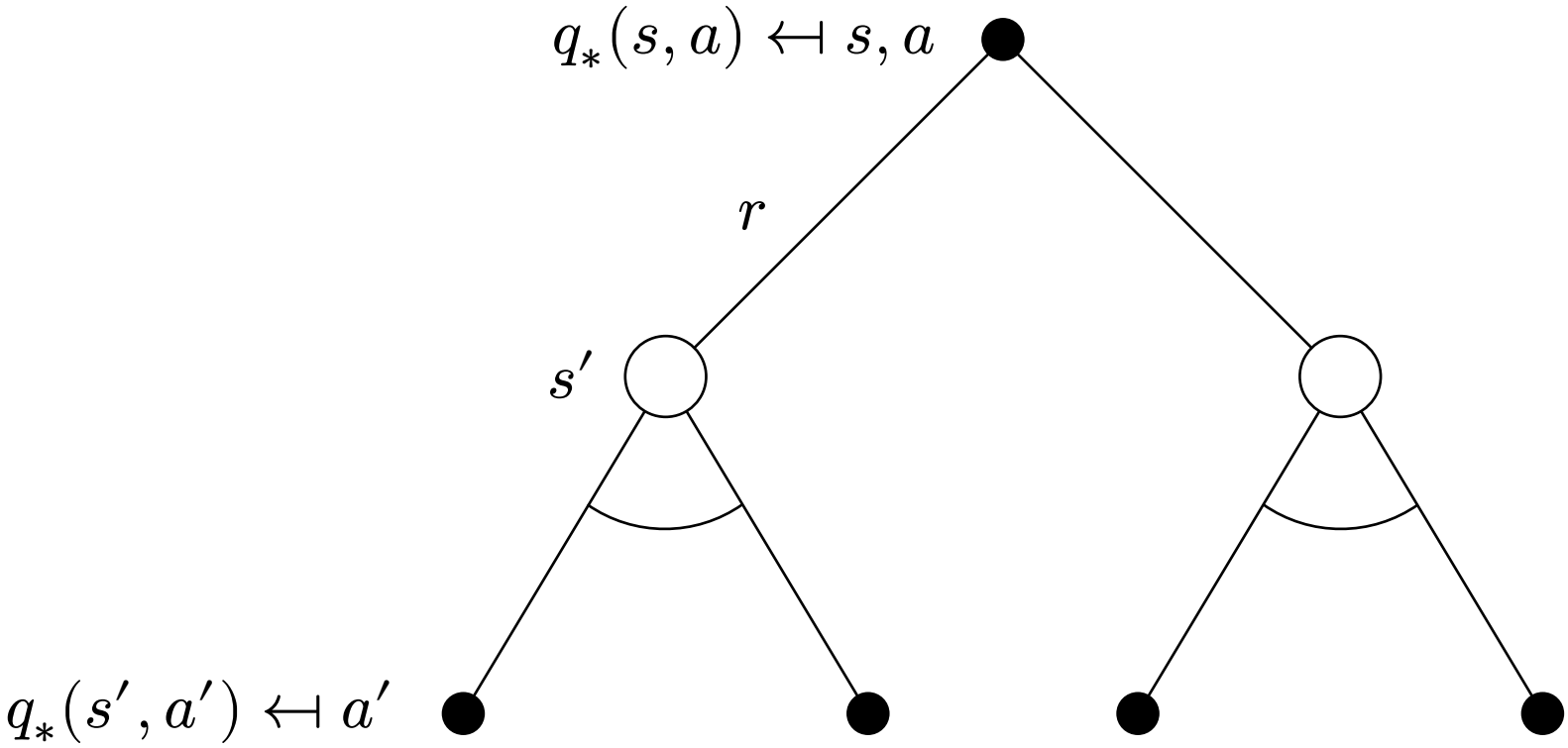
Proposition 3.9.2

$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \max_{a'} q_*(s', a')$$

3.10 Backup diagrams



3.10 Backup diagrams



3.11 Onto the practice session

From the Bellman equations to:

- Bellman equation for the *state*-value function $\Longrightarrow$ Value iteration
- Bellman equation for the *action*-value function $\Longrightarrow$ Q-learning